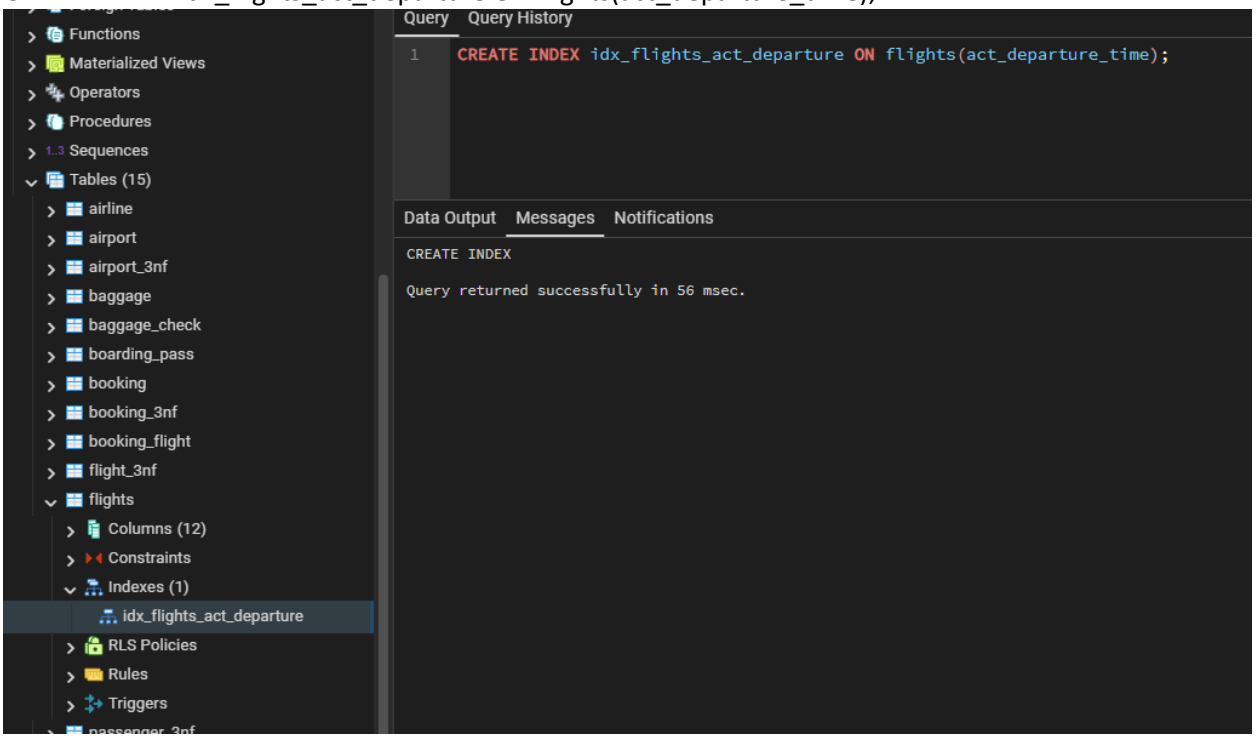
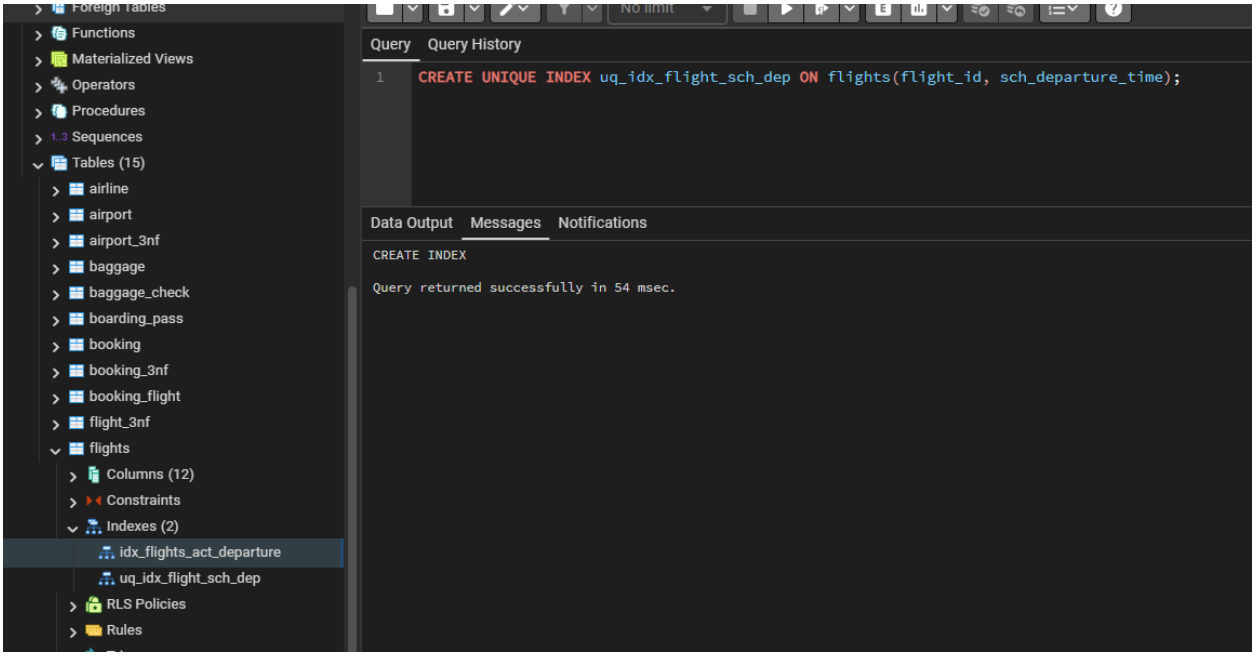
Lab 7

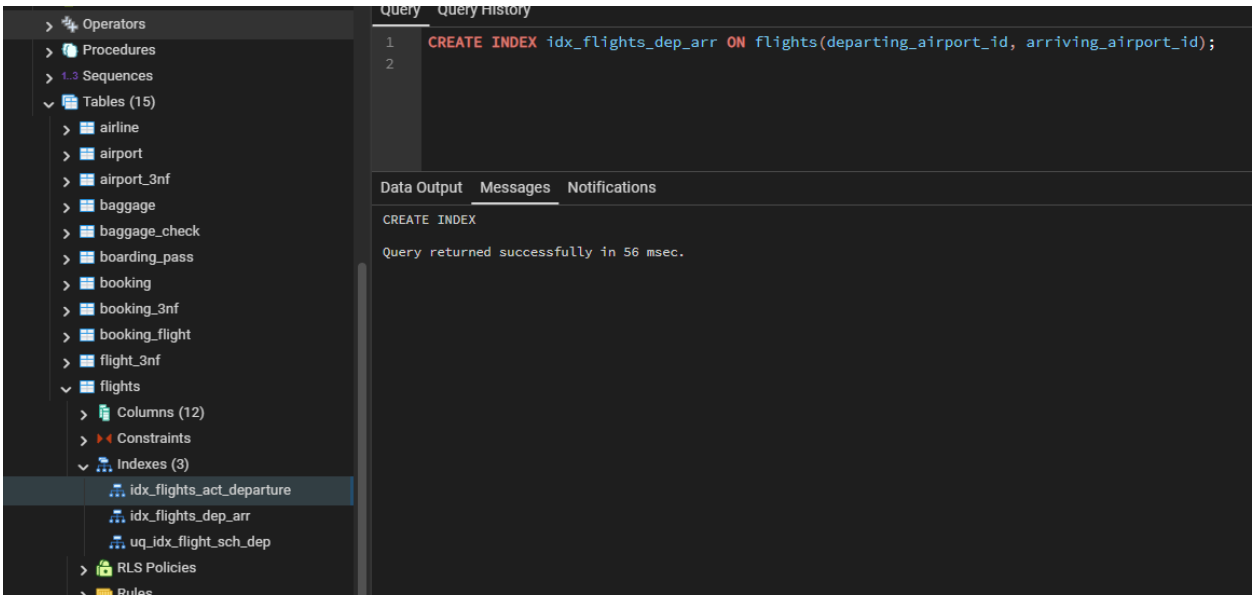
1. CREATE INDEX idx_flights_act_departure ON flights(act_departure_time);



2. CREATE UNIQUE INDEX uq_idx_flight_sch_dep ON flights(flight_id, sch_departure_time);



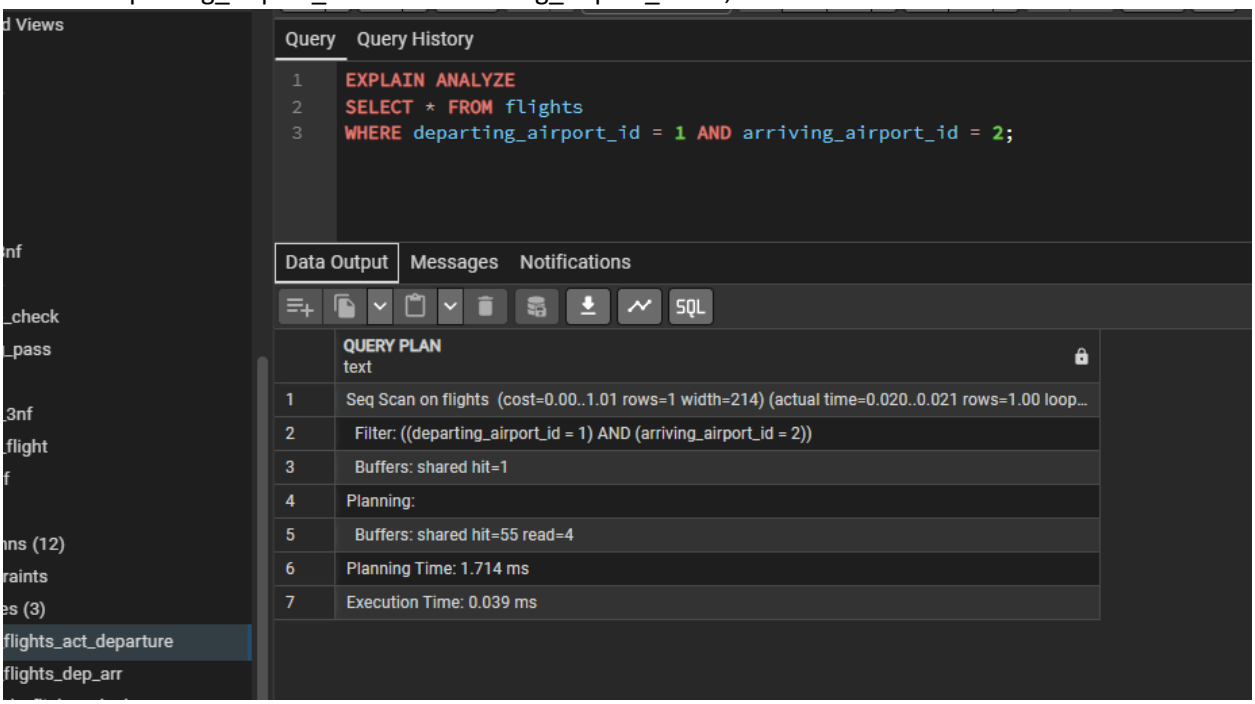
3. CREATE INDEX idx_flights_dep_arr ON flights(departing_airport_id, arriving_airport_id);



4&5. EXPLAIN ANALYZE

SELECT * FROM flights

WHERE departing_airport_id = 1 AND arriving_airport_id = 2;



6. CREATE UNIQUE INDEX uq_passengers_passport ON passengers(passport_number);

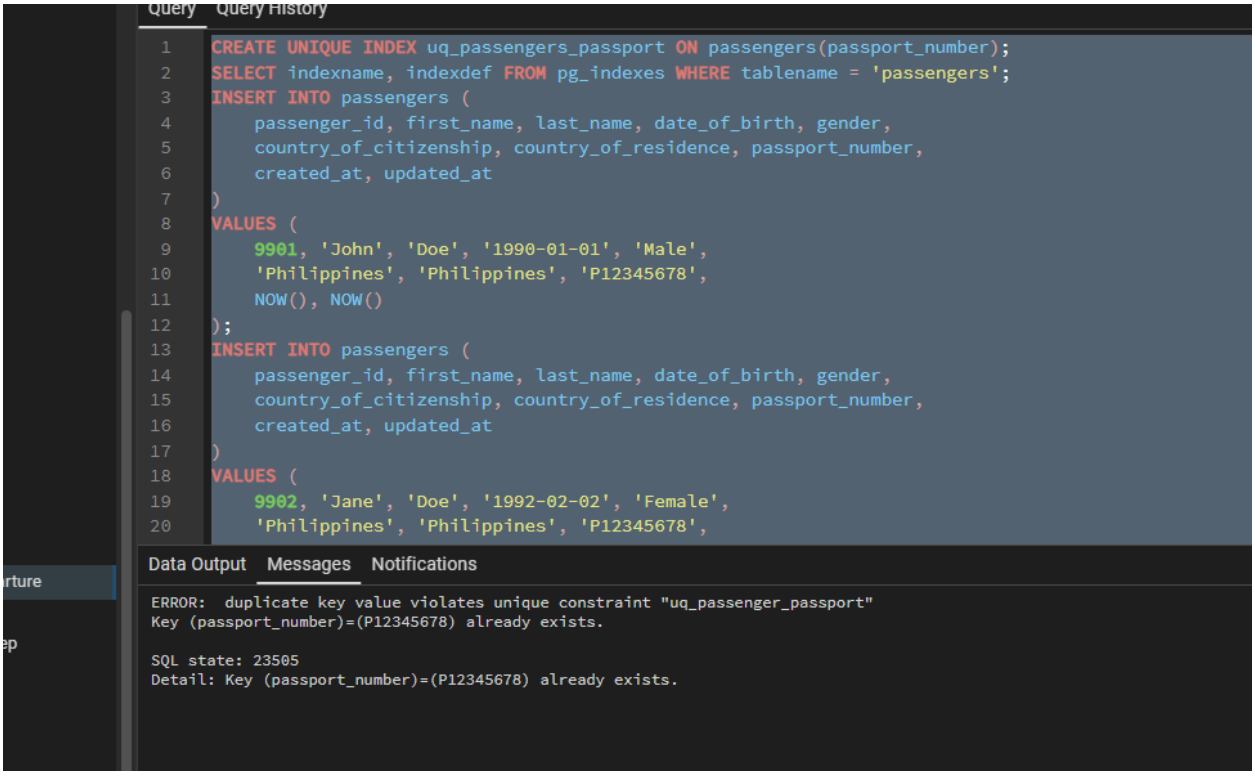
SELECT indexname, indexdef FROM pg_indexes WHERE tablename = 'passengers';

INSERT INTO passengers (

passenger_id, first_name, last_name, date_of_birth, gender,

country_of_citizenship, country_of_residence, passport_number,

```
    created_at, updated_at
)
VALUES (
    9901, 'John', 'Doe', '1990-01-01', 'Male',
    'Philippines', 'Philippines', 'P12345678',
    NOW(), NOW()
);
INSERT INTO passengers (
    passenger_id, first_name, last_name, date_of_birth, gender,
    country_of_citizenship, country_of_residence, passport_number,
    created_at, updated_at
)
VALUES (
    9902, 'Jane', 'Doe', '1992-02-02', 'Female',
    'Philippines', 'Philippines', 'P12345678',
    NOW(), NOW()
);
```



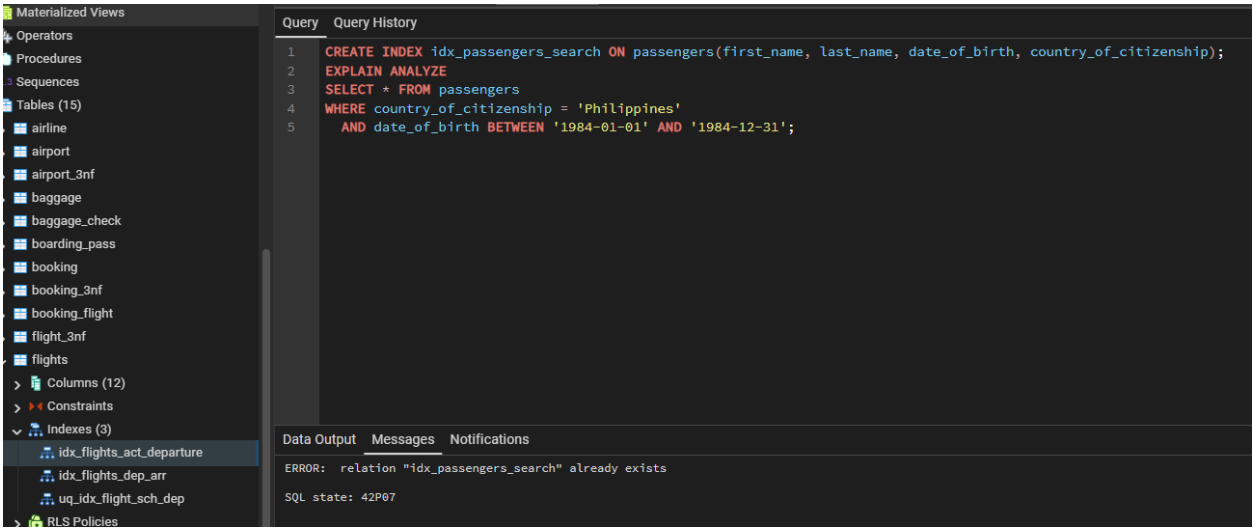
7. CREATE INDEX idx_passengers_search ON passengers(first_name, last_name, date_of_birth, country_of_citizenship);

EXPLAIN ANALYZE

SELECT * FROM passengers

WHERE country_of_citizenship = 'Philippines'

AND date_of_birth BETWEEN '1984-01-01' AND '1984-12-31';



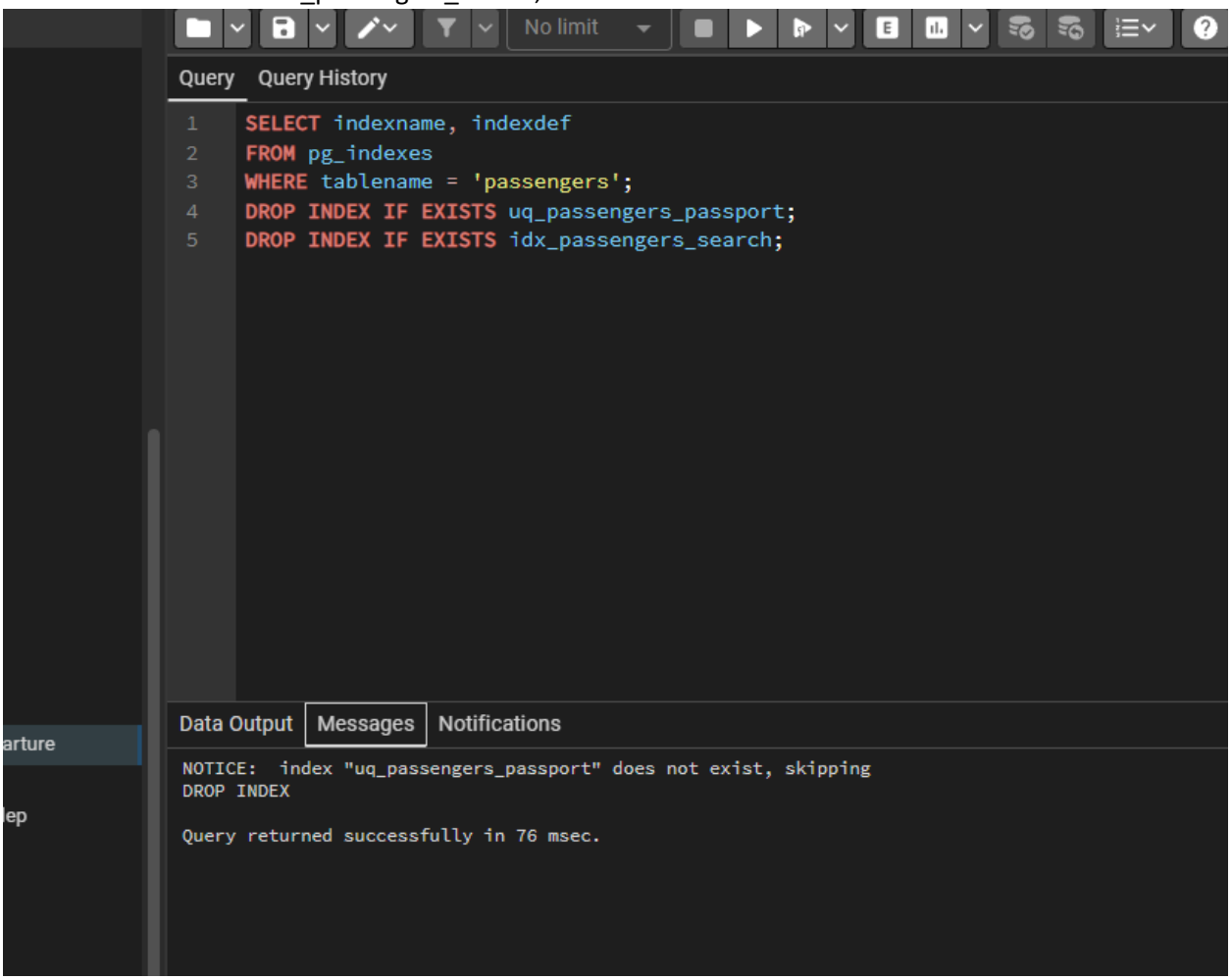
8. SELECT indexname, indexdef

FROM pg_indexes

WHERE tablename = 'passengers';

DROP INDEX IF EXISTS uq_passengers_passport;

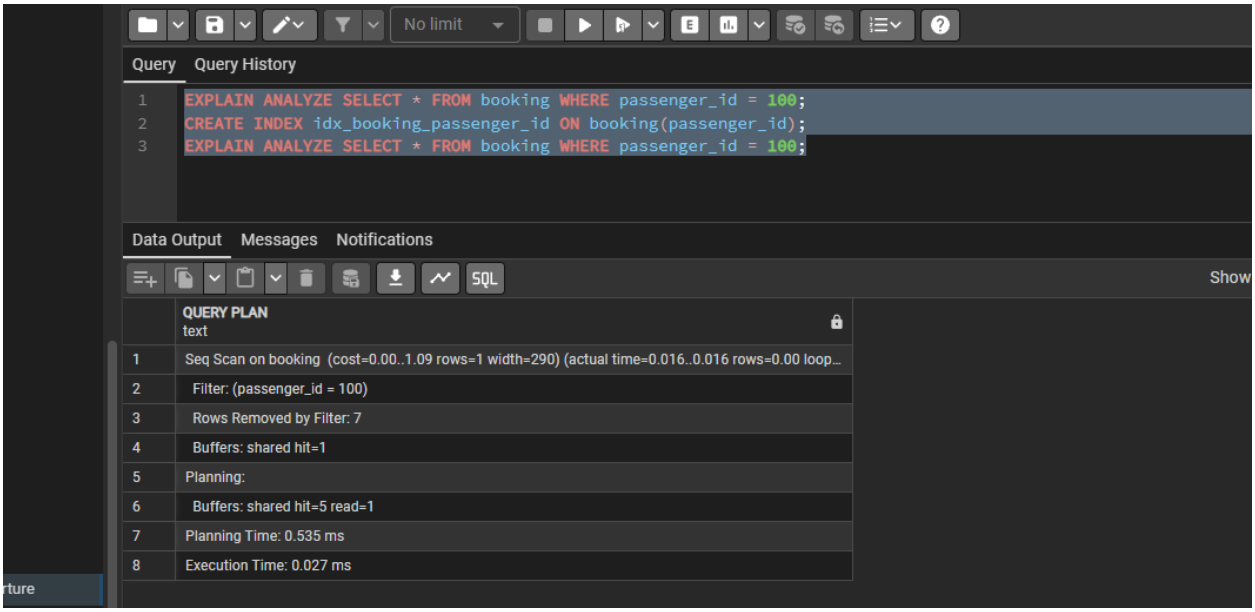
DROP INDEX IF EXISTS idx_passengers_search;



9. EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;

CREATE INDEX idx_booking_passenger_id ON booking(passenger_id);

EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;



10. CREATE INDEX idx_booking_flight_ids ON booking_flight(booking_id, flight_id);

EXPLAIN ANALYZE

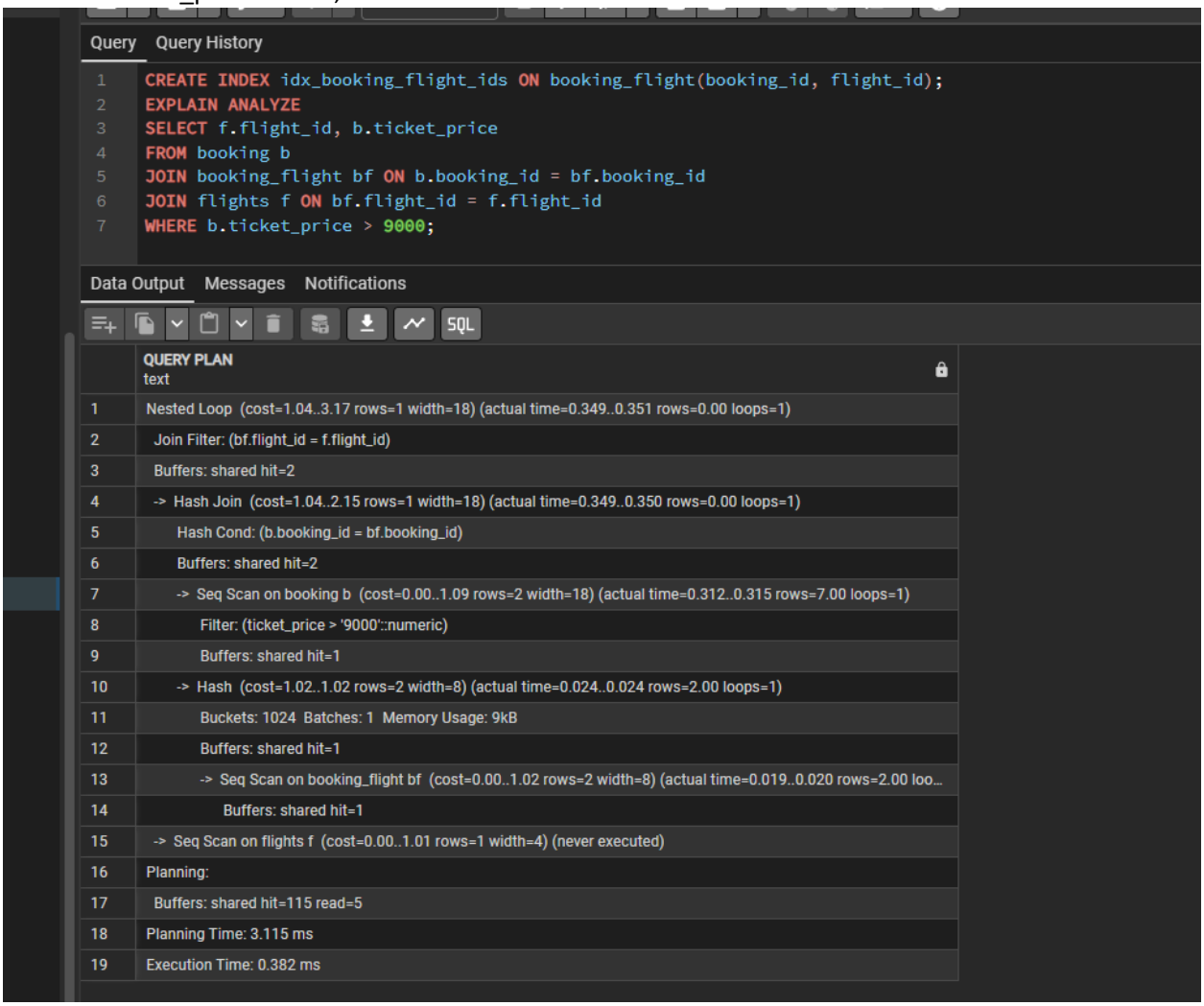
SELECT f.flight_id, b.ticket_price

FROM booking b

JOIN booking_flight bf ON b.booking_id = bf.booking_id

JOIN flights f ON bf.flight_id = f.flight_id

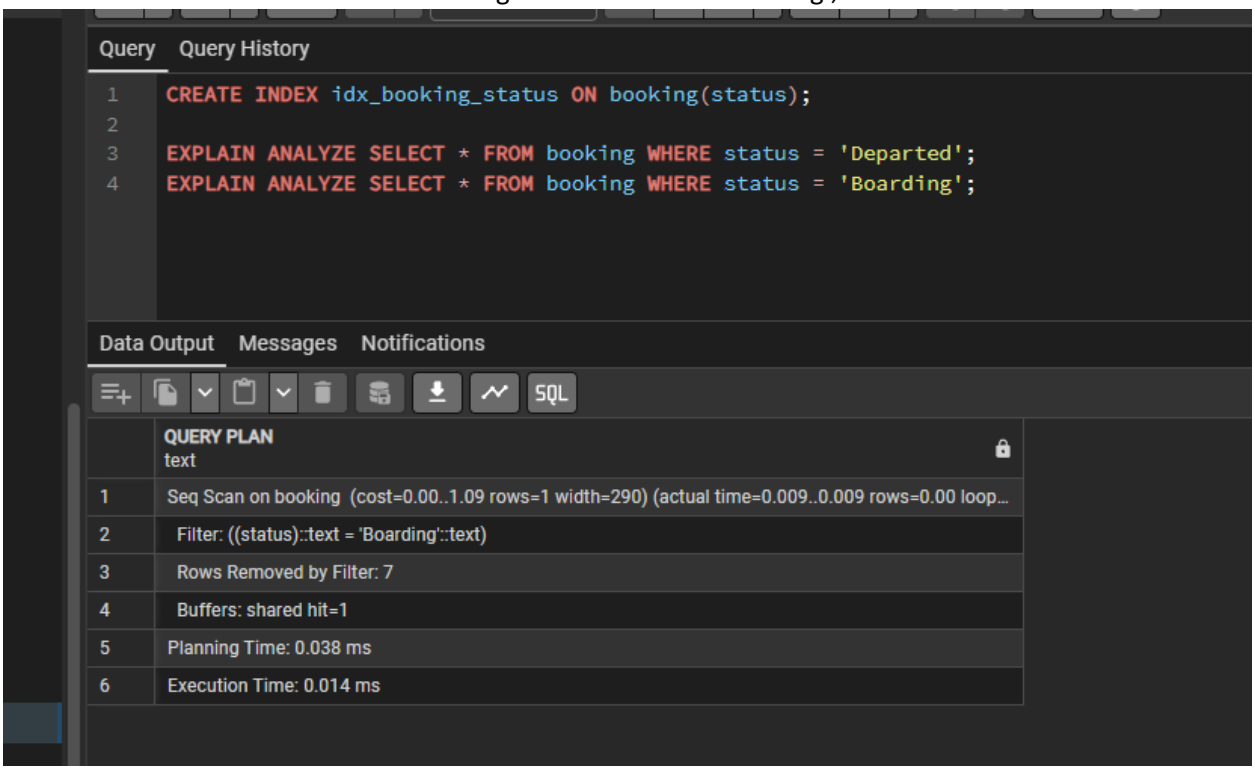
WHERE b.ticket_price > 9000;



11. CREATE INDEX idx_booking_status ON booking(status);

EXPLAIN ANALYZE SELECT * FROM booking WHERE status = 'Departed';

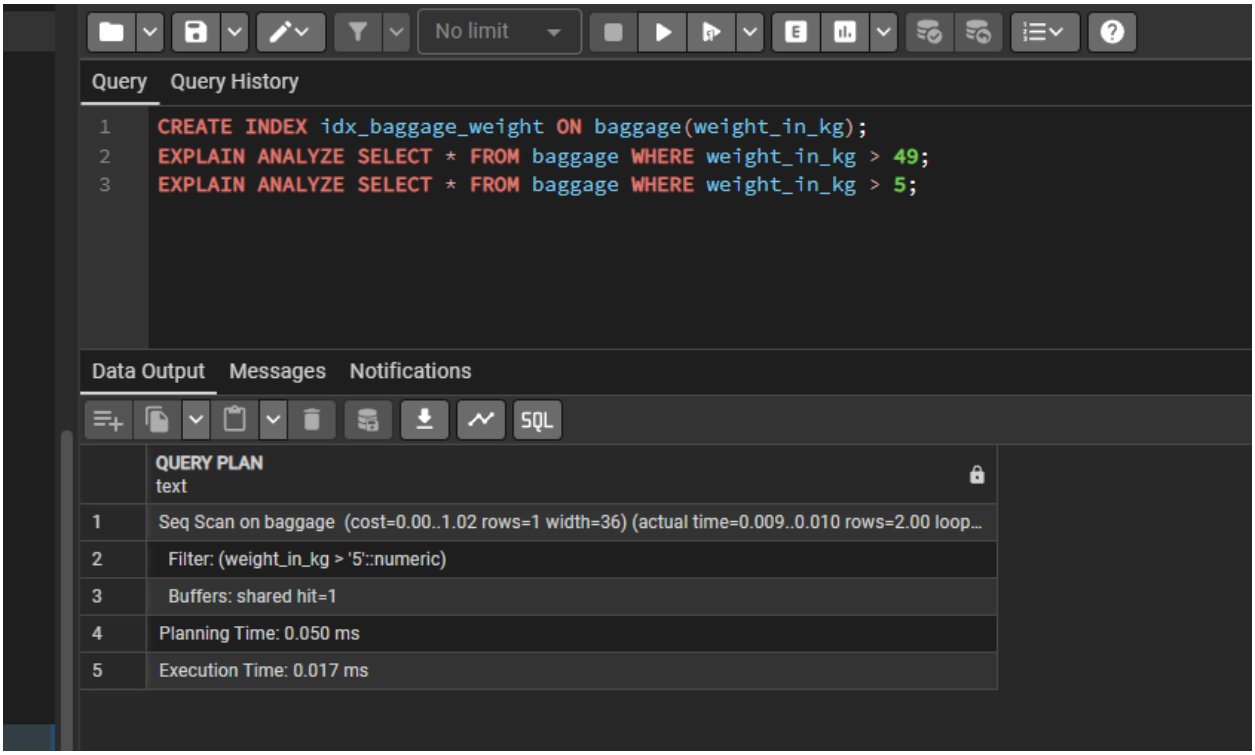
EXPLAIN ANALYZE SELECT * FROM booking WHERE status = 'Boarding';



12. CREATE INDEX idx_baggage_weight ON baggage(weight_in_kg);

EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 49;

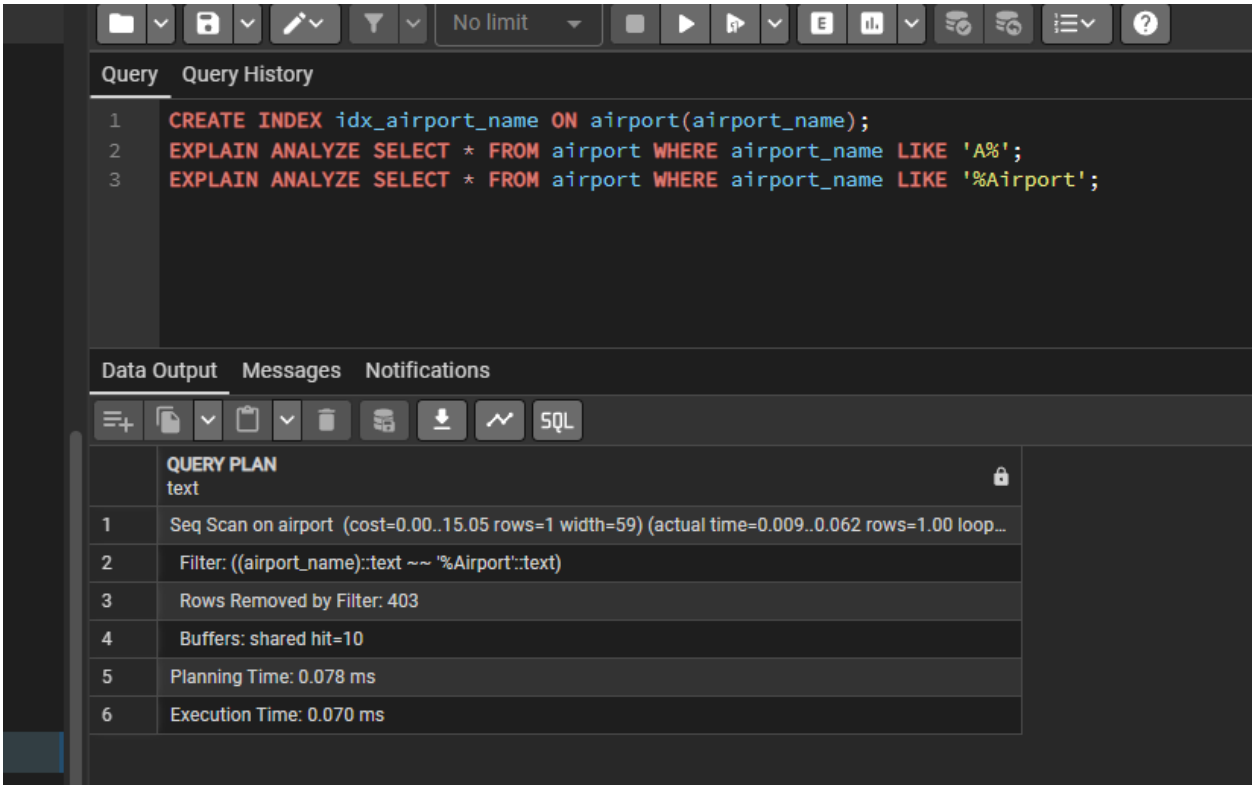
EXPLAIN ANALYZE SELECT * FROM baggage WHERE weight_in_kg > 5;



13. CREATE INDEX idx_airport_name ON airport(airport_name);

EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE 'A%';

EXPLAIN ANALYZE SELECT * FROM airport WHERE airport_name LIKE '%Airport';

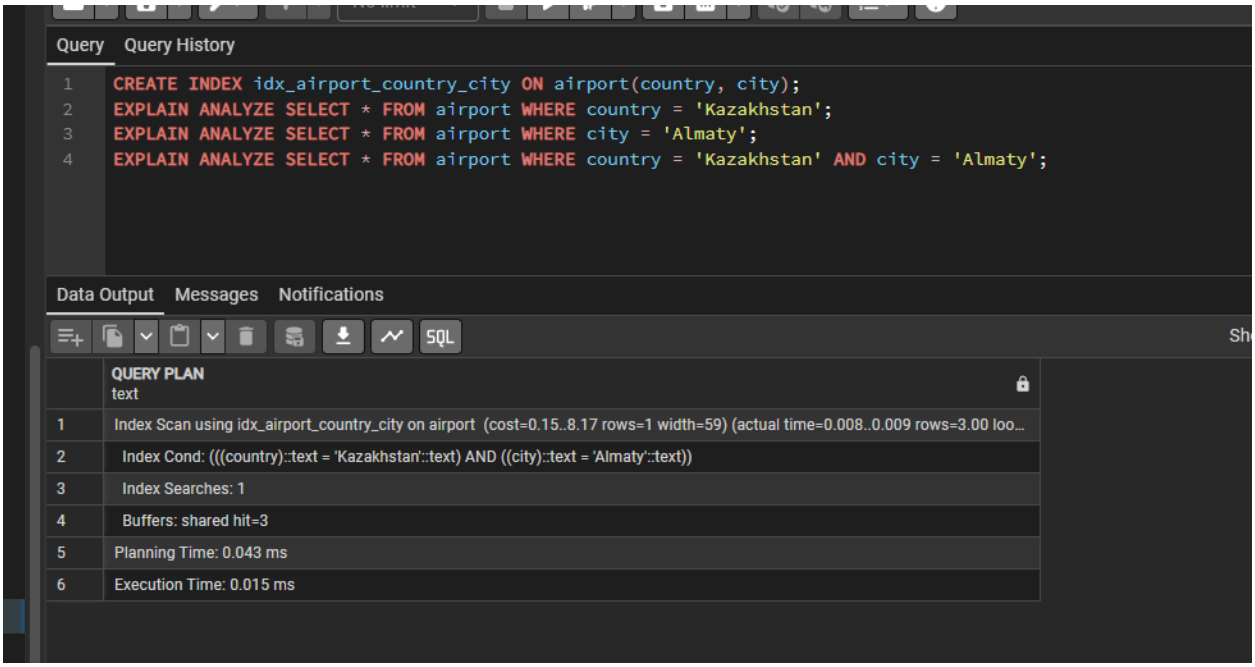


14. CREATE INDEX idx_airport_country_city ON airport(country, city);

EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Kazakhstan';

EXPLAIN ANALYZE SELECT * FROM airport WHERE city = 'Almaty';

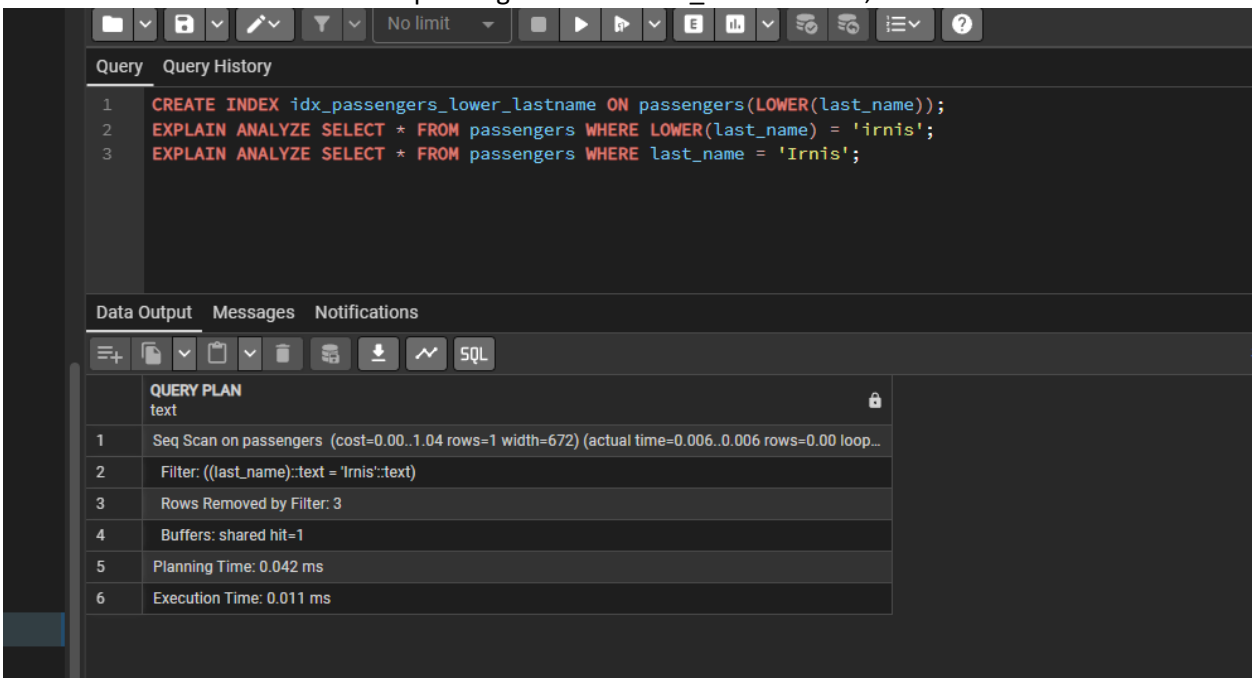
EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Kazakhstan' AND city = 'Almaty';



15. CREATE INDEX idx_passengers_lower_lastname ON passengers(LOWER(last_name));

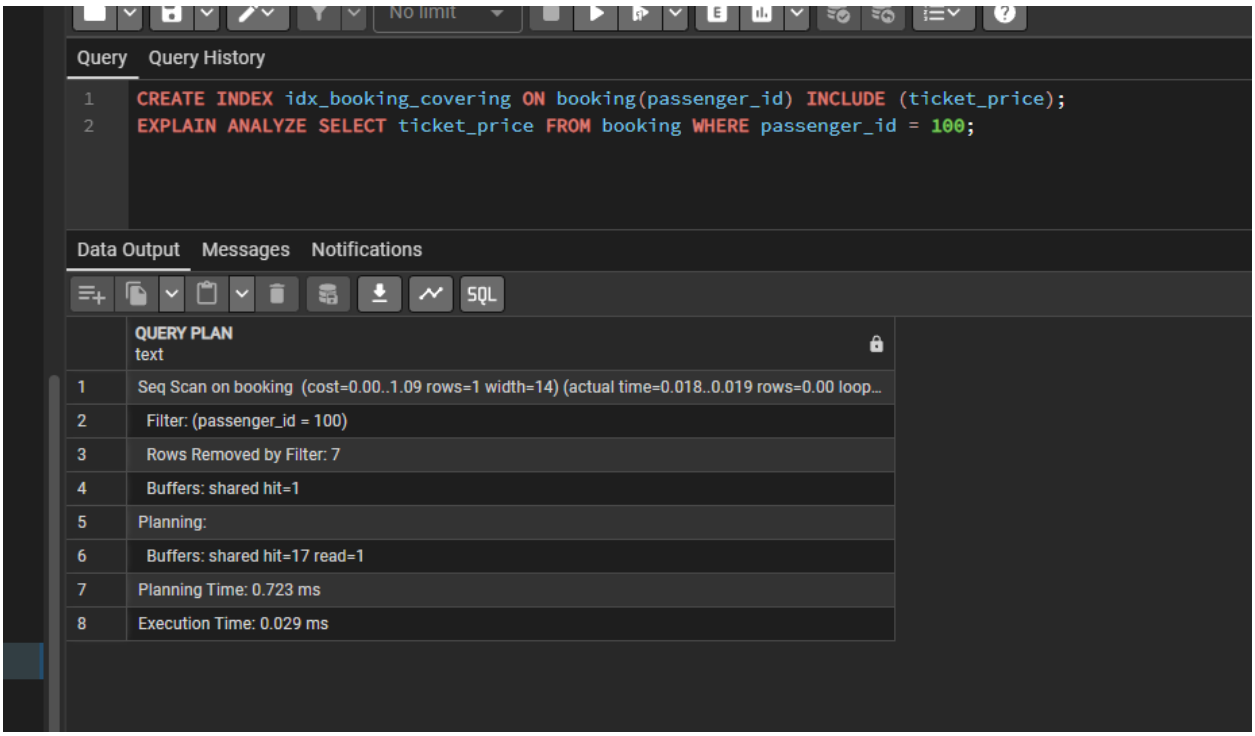
EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';

EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'Irnis';



16. CREATE INDEX idx_booking_covering ON booking(passenger_id) INCLUDE (ticket_price);

EXPLAIN ANALYZE SELECT ticket_price FROM booking WHERE passenger_id = 100;



17. SELECT airline_code, COUNT(*)

FROM airline

GROUP BY airline_code

HAVING COUNT(*) > 1;

CREATE UNIQUE INDEX uq_airline_code ON airline(airline_code);

